Global Supply Chain Analytics

EDA & Business Intelligence Report | March 2026

10,000
Total Orders 📦

87.1%
On-Time Delivery ✅

$13,833
Avg Cost (High-Risk) 💰

8.8%
Major Delay Rate ⚠️

EDA Insights

Exploratory Data Analysis findings from 10,000 shipment records

📋 1. Missing Data Alert

- **8,733 missing values** found in `Disruption_Event` column
- Critical data gap requires imputation or flagging before predictive modeling
- Suggests poor tracking of minor disruptions

🌍 2. Export Origins — Balanced Operations

- **Top Performer:** Mumbai, India (1,710 orders)

- Closely followed by Shenzhen (1,702) and Hamburg (1,699)
- **High Consistency:** Major hubs range between 1,600–1,710 orders
- ⚠️ **New York Outlier:** Only 2 orders — likely data error or niche lane

🏭 3. Import Destinations — Western Hubs

- **Top:** Felixstowe, UK (1,710) and New York, US (1,699)
- Rotterdam, NL (1,702) follows closely
- **Hamburg Anomaly:** Top exporter but only 2 import orders
- Strong demand confirmed in Western markets (US/EU)

🚢 4. Transport Mode — Sea Dominates

- **Sea:** 8,271 orders (>80%) — backbone of operations
- **Air:** 1,729 orders (~17%) — premium/urgent segment
- Sea-to-Air ratio ≈ 5:1 indicating bulk shipping focus

⏱️ 5. Lead Time Distribution

- **Most frequent:** 24 days (1,445 orders)
- Bulk concentrated in 7–35 day range
- **Express Segment:** ~1,700 orders with 2–4 day lead times
- Clear split: Air (2–3 days) vs Sea (7+ days)

📅 6. Planned vs. Actual — Bi-Modal

- **Primary Peak:** 3–7 days (Air freight/Fast lane)
- **Secondary Peak:** ~25 days (Standard Sea freight)
- **Lull Period:** 20–23 days rarely used as targets

📉 7. Actual Delivery Reality

- Deliveries stretch beyond 35-day plan, up to **50 days**
- **Bottleneck:** Drop at 15 days, surge at 18–25 days
- Sea shipments most prone to unpredictable long-tail delays

✅ 8. Delay Distribution

- **90% Punctuality:** ~9,000 orders have < 2.5 days delay
- **Long Tail:** 500+ orders delayed 5–20 days
- Risk: If not delivered within 48h, likely to face 10+ day delay

🏆 9. On-Time Delivery Rate

- **87.1% Success Rate:** 8,710 orders on time
- **12.9% Failure Rate:** 1,290 orders Late
- Roughly 1 in 8 orders misses the deadline

⚠️ 10. Disruption Root Causes

- **Port Congestion:** 573 orders (Most Frequent)
- **Geopolitical Conflict:** 521 orders (Most Severe)
- **Severe Weather:** 173 orders (Lowest Impact)
- Combined causes account for ~86% of all delays

🌐 11. Geopolitical Risk Exposure

- Risk uniformly spread from 0.1 (low) to 0.9 (high)
- Consistent 400–500 orders per bin even at high risk
- **No "Zero-Risk"** operations exist in the network

12. Risk Category Breakdown

- **Medium Risk:** 50% (5,004 orders)
- **Low Risk:** 26.2% (2,622 orders)
- **High Risk:** 23.7% (2,374 orders)
- 24% of volume moves through high-tension zones

13. Weather Impact

- **Moderate Weather:** 293 orders (Most common hurdle)
- **Normal Conditions:** 279 orders still faced delays
- **Severe Weather:** 186 orders (Least frequent but impactful)

14. Inflation Environment

- **Normal Inflation:** 78% (7,784 orders) — Stable
- **High Inflation:** 11% (1,113 orders) — Margin pressure
- Cost fluctuations are cyclical, not aggressive trends

15. Shipping Cost Distribution

- **92% of orders** cost < $50,000 (Cost-Optimized)
- 600–700 orders in $50k–$100k range (Bulk/Air)
- **Outliers:** <1% exceed $150k–$200k
- Budget is highly predictable with few spikes

16. Cargo Weight Profile

- **Flat Distribution:** 100 kg to 10,000 kg
- Slight peaks at 6,000 kg & 8,000 kg (FCL loads)
- Consistent mid-range flow (500–5,000 kg)
- Full capacity utilization across fleet

17. Mitigation Strategy

- **Standard Shipping:** 93% (9,283 orders) — Passive
- **Re-routing:** 3.7% (368 orders)
- **Expedited Air:** 3.5% (349 orders)
- Strategy favors margin protection over speed recovery

Key Business Insights

Financial and operational impact analysis

Risk vs. Cost

- High-risk zones cost **~54% more** ($13,833 vs $8,966)
- Each risk level adds ~$2,400 "risk surcharge"
- High-risk costs are the most volatile

Most Expensive Events

- **Geopolitical:** $23,023 avg per shipment
- **Port Congestion:** $15,530 avg
- **Severe Weather:** $12,530 avg
- Conflict costs nearly 2x more than weather

Total Financial Loss

- **Geopolitical:** ~$12M total loss
- **Port Congestion:** ~$8.8M total loss
- ⚠️ **Unknown:** $91.3M (Untracked risk)

Efficiency (Cost/kg)

- **Geopolitical:** $4.47/kg (Least efficient)
- **Port Congestion:** $3.06/kg
- **Severe Weather:** $2.68/kg

- Weather impact is lowest ($2.1M)
- Conflict causes 118% price hike vs baseline

Delay Analysis

- Avg delay ~0.95 days across all risk levels
- Max delay of 20 days possible in ALL zones
- High-risk zones have more "secondary peak" (15-20 day) delays

Transport Mode Stats

- **Air:** 0.92 days avg delay
- **Sea:** 0.95 days avg delay
- Sea has 741 high-delay cases vs Air's 141
- Reliability gap is small (8% vs 9%)

High-Delay Cost Impact

- **>3 Days Delay:** $19,077 avg cost
- **≤3 Days Delay:** $10,699 avg cost
- Major delay = **78% cost increase**
- 8.8% of orders drive disproportionate loss

Mitigation Costs

- **Standard:** $10,082 (Baseline)
- **Re-routing:** $6,078 (Cheapest)
- **Air Freight:** $53,130 (Most Expensive)
- Air saves ~5 days but costs $47k extra

Forecasting Outlook

- **Window:** Feb–May 2026 (Stability)
- **Predicted Delay:** ~0.95 days
- No major spikes anticipated in Q2
- Ideal time for route optimization

Integrated Supply Chain Summary

Strategic overview of key logistics metrics

CATEGORY	KEY METRIC / FINDING	STRATEGIC BUSINESS INSIGHT
Logistics Spine	Mumbai (Origin) → Felixstowe (Destination)	Primary trade corridor; any disruption here has the largest impact on total revenue
Reliability	87.1% On-Time Delivery	Strong performance baseline, yet 1,290 orders require bottleneck analysis
Primary Disruptor	Port Congestion (573 cases)	Most frequent cause of delay, typically resulting in 2–4 day setbacks
Highest Risk	Geopolitical Conflict (521 cases)	Most severe threat, causing catastrophic delays ranging from 10–16 days
Financial Impact	54% Cost Premium in High-Risk Zones	Moving goods through volatile areas costs $13,833 avg — significantly higher
Environmental	Moderate Weather is top factor	Supply chain is sensitive to standard seasonal shifts, not just extreme disasters
Cost Efficiency	92% of orders < $50,000	Most shipping is cost-optimized; high spend does not guarantee faster delivery
Cargo Profile	Uniform Weight (100kg - 10,000kg)	High operational versatility with no dependency on specific cargo size

CATEGORY	KEY METRIC / FINDING	STRATEGIC BUSINESS INSIGHT
Risk Response	93% Standard Shipping mitigation	Strategy favors cost preservation over speed when disruptions occur
Economic State	78% Normal Inflation level	Operations stable but sensitive to global fuel or labor spikes

Actionable Recommendations

Strategic initiatives based on data findings

1

Diversify Routes

- Develop **"Plan B" routes** for the Mumbai-Felixstowe corridor
- Port Congestion and Geopolitical Conflict are primary threats
- Protect the 13% of orders falling behind schedule
- Pre-negotiate alternative port agreements

2

Re-evaluate Premium Shipping

- High costs do **NOT** correlate with fewer delay days
- Audit high-cost carrier contracts for service compliance
- Redirect savings from overpriced carriers to mitigation tools
- Benchmark carrier performance per route

3

Shift to Active Mitigation

- Reduce 93% reliance on passive Standard Shipping
- Use **Re-routing** aggressively to prevent 20-day delays

- Reserve Expedited Air only for critical high-value cargo
- Implement real-time disruption dashboards

ML Model Performance

Predictive analytics and regression results

Best Model

Algorithm	**Random Forest**
R^2 Score	**0.996**
MAE	**$247**
Status	✅ **Top Performer**

Time Series

Model	**SARIMAX**
Validation	**Diebold-Mariano**
p-value	**0.015**
Result	**Superior to ARIMA**

Statistical Regression

Weight Coeff	**1.0187 (1:1)**
Delay Tax	**+4.1% / day**
R^2 Score	**0.483**
Insight	**Weight drives Cost**